Vedantu
YOUR PERSONAL TEACHER ONLINE

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s-block

1. ALKALI METALS (GROUP 1)

The group 1 elements have ns^1 electronic configuration and are highly reactive metals

Elements	Atomic Number	Electronic Configuration
Lithium (Li)	3	$[He]\ 2s^1$
Sodium (Na)	11	$[Ne]\ 3s^1$
Potassium (K)	19	$[Ar]\ 4s^1$
Rubidium (Rb)	37	$[Kr]\ 5s^1$
Cesium (Cs)	55	$[Xe]\ 6s^1$
Francium (Fr)	87	$[Rn]\ 7s^1$

1.1 Physical Properties

1.1.1 Atomic Size

The atoms are largest in their corresponding periods. Atomic size increases in going down the group.

1.1.2 Oxidation State

The group 1 elements exhibit +1 oxidation state

1.1.3 Density

Alkali metals have large size which accounts for their low density

$$\text{Density} = \frac{\text{Atomic mass}}{\text{Atomic volume}}$$

Atomic weight increases from Li to Cs in the group and volume also increases but increase in atomic weight is more than increase in volume. Therefore density increases from Li to Cs.

Exception : Density of Na is more than that of K

order : Li < K < Na < Rb < Cs

1.1.4 Nature of Bonds

The electronegativity values being low, they combine with other elements to form Ionic bond.

1.1.5 Ionization Energy

The first ionisation energies for the atoms in this group are lower than those for any other group in the periodic table. The atoms are very large so the outer electron are held weakly by the nucleus hence the ionisation energy is not large. Ionization energy decreases on moving down the group.

1.1.6 Flame Test

Alkali metals have large size, when heated on the flame the electrons present in the valence shell move from lower energy level to higher energy level by absorption of heat from the flame. When they come back to the ground state, they emit the extra energy in the form of visible light to provide colour to the flame

Element	Colour
Li	Red
Na	Golden yellow
K	Violet
Rb	Red Violet
Cs	Blue

1.1.7 Standard Oxidation Potential

The measure of the tendency of donating electrons of a metal in water is called its electrode potential. If the concentration of metal ions is unity, then it is called as standard electrode potential. Lithium has the highest ionization potential but has an highest electrode potential due to highest hydration energy.

1.1.8 Hydration of Ions

The ions are heavily hydrated. The smaller the size of the ion, the greater is the degree of hydration. Thus the degree of hydration decreases down the group from Li^+ to Cs^+. Thus with the increase in hydration electrical conductivity decreases.

1.1.9 Lattice Energy

Salts of alkali metals are ionic solids. Lattice energy of salts of alkali metals having common anion decreases on descending down the group.

1.1.10 Solubility in Liquid Ammonia

$M + n\,NH_3 \rightarrow [M(NH_3)_x]^+ + e^-(NH_3)_y$

$(n = x + y)$

Dilute solutions of alkali metals in liquid ammonia are dark blue in colour and the main species present are solvated metal ions and solvated electrons. If the blue solution is allowed to stand, the colour fades until it dissappears owing to the formation of metal amide. The solutions of metal in liquid conduct electricity because of the presence of solvated electrons. The dilute solutions are paramagnetic because they contain free electrons.

1.1.11 Electronegativity Values

The electronegativity values are small which decrease from Li to Cs.

1.1.12 Reactivity

The reactivity of alkali metals goes on increasing in the following order.

$Li < Na < K < Rb < Cs$

1.1.13 Colourless and Diamagnetic ions

The property of an ion being colourless or coloured depends on the number of unpaired electrons present in the ion. If unpaired electrons are present in anion then these electrons can be excited by energy from light and show colour on coming back to the ground state. The ion which have unpaired electrons show magnetic properties whereas the ions having paired electrons nullify the magnetic fields of each other. Such ions are called diamagnetic ions.

Super oxides are para magnetic and coloured due to the presence of unpaired electrons.

1.1.14 Melting and Boiling Point

The cohesive energy is the force holding the atoms or ions together in the solid. The cohesive energy depends on the number of electrons that can participate in bonding. The cohesive force decreases down the group in alkali metals group as they have only one valence electron which participates in bonding and of the large size and diffusing nature of the outer bonding electron. The atoms become larger on descending down the group, so the bonds are weaker, the cohesive energy decreases and the softness of metal increases. Hence the melting point decreases down the group. Boiling point also decreases down the group.

1.2 Chemical Properties

Some common reactions of Group 1 metals

Reaction	Comment
$M + H_2O \rightarrow MOH + H_2$	Hydroxides are strongest base known
$Li + O_2 \rightarrow Li_2O$	Monoxide formed by Li and to a small extent by Na
$Na + O_2 \rightarrow Na_2O_2$	Peroxide formed by Na and to a small extent by Li
$K + O_2 \rightarrow KO_2$	Superoxide formed by K, Rb, Cs.
$M + H_2 \rightarrow MH$	Ionic salt like hydrides
$Li + N_2 \rightarrow Li_3N$	Nitride formed only by Li
$M + S \rightarrow M_2S$	All metals form sulphides
$M + X_2 \rightarrow MX$	All the meatls form halide
$M + NH_3 \rightarrow MNH_2$	All the metals form amides.

1.2.1 Reaction with Air

Group 1 elements are very reactive and tarnish rapidly in air. Cs burns spontaneously in air.

These metals form alkaline carbonates in moist air.

$2Na + O_2 \rightarrow 2\,Na_2O$

$Na_2O + H_2O\,(\text{mositure}) \rightarrow 2\,NaOH$

$2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$

1.2.2 Reaction with O_2

Li forms Li_2O, Na forms two type of oxide

(M_2O, M_2O_2) and K, Rb, Cs forms superoxides (MO_2).

1.2.2.1 Basic Nature, Ionic Nature of the Oxides

Basic nature of oxides increases from Li to Cs due to increase in the size of cation

- Size of cation increases from Li to Cs. According to Fajan's Rule, ionic character of these oxides increases from Li to Cs.
- Solubility in water increases from Li to Cs oxides, due to increase in ionic character of these metal oxides.

1.2.3 Reaction with water

Group 1 metals react with water liberating H_2 and forming hydroxides.

$2\,Li + 2\,H_2O \rightarrow 2LiOH + H_2$

$2Na + 2\,H_2O \rightarrow 2NaOH + H_2$

$2K + 2\,H_2O \rightarrow 2\,KOH + H_2$

1.2.4 Reaction with Hydrogen

Group 1 metals reacts with H_2 to form ionic hydrides. Thermal stability of LiH is high.

Stability of hydrides :

$LiH > NaH > KH > RbH > CsH$

1.2.5 Reaction with dilute acids

Due to alkaline nature, these metals react rapidly with dilute acids and the rate of reaction increases from Li to Cs because of increase in basic character.

1.3 Compounds of Alkali Metal

1.3.1 Hydroxides

NaOH is often called as caustic soda. KOH is called caustic potash because of their corrosive properties. The caustic alkali are the strongest base in aqueous solution. The solubility of hydroxides increases down the group.

The bases react with acids to form salt and H_2O.

$KOH + HCl \rightarrow KCl + H_2O$

$NaOH + HCl \rightarrow NaCl + H_2O$

$2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$

The bases liberate ammonia from ammonium salts

$NaOH + NH_4Cl \rightarrow NH_3 + NaCl + H_2O$

$KOH + NH_4Cl \rightarrow NH_3 + KCl + H_2O$

KOH resembles NaOH in all its reaction. but as KOH is much more expensive it is seldom used. However KOH is more soluble in alcohol, thus producing $C_2H_5O^-$ ions by the equilibrium

$C_2H_5OH + OH^- \rightleftharpoons C_2H_5O^- + H_2O$

This accounts for the use of alcoholic KOH in organic chemistry Group 1 hydroxides are thermally stable.

1.3.2 Oxides, Peroxides, Superoxides

Normal oxides - monoxide : The monoxides are ionic. They are strongly basic oxides and they react with water form strong bases.

$Na_2O + H_2O \rightarrow NaOH$

$K_2O + H_2O \rightarrow KOH$

Peroxides

Preparation :

$2Na + O_2\text{ (excess)} \xrightarrow{300°C} Na_2O_2$

$2Na_2O \xrightarrow{7400°C} Na_2O_2 + Na\text{ (vapour)}$

Properties

$Na_2O_2 + H_2SO_4\text{ (dil)} \rightarrow Na_2SO_4 + H_2O_2$

$Na_2O_2 + 2H_2O \rightarrow 2NaOH + H_2O_2$

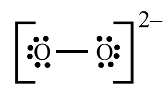
Na_2O_2 is a powerful oxidant Because it reacts with CO_2 in the air it has been used to purify the air in submarines

$Na_2O_2 + CO \rightarrow Na_2CO_3$

$Na_2O_2 + 2CO_2 \rightarrow Na_2CO_3 + O_2$

$Na_2O_2 + Cr^{3+} \rightarrow CrO_4^{2-}$

Structure



peroxide ion

O is sp^3 hydridised. The peroxide ion has 18 electrons which occupy the molecular orbitals as shown.

$\sigma 1s^2, \sigma^*1s^2, \sigma 2s^2, \sigma^*2s^2, \sigma 2p_z^2, \pi 2p_y^2 = \pi 2p_z^2, \pi^*2p_y^2 = \pi^*2P_z^1$

Bond order is 1 and it is diamagnetic

Superoxides (O_2^-)

Superoxides are ionic oxides $M^+O_2^-$

Preparation :

$M + O_2$ (excess) $\rightarrow MO_2$

(M = K, Rb, Cs)

Superoxides are stronger oxidizing agents than peroxides. The stability of these superoxides is in the order

$KO_2 < RbO_2 < CsO_2$

Reactions

$KO_2 + H_2O \rightarrow KOH + H_2O_2 + 1/2\ O_2$

KO_2 is used in space capsules, submarines and breathing masks because it produces O_2 and removes CO_2

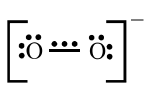
$4KO_2 + 2CO_2 \rightarrow 2\ K_2CO_3 + 3O_2$

$4KO_2 + 4CO_2 + 2\ H_2O \rightarrow 4KHCO_3 + O_2$

sodium superoxide cannot be prepared by burning metal in oxygen but it can be prepared by reacting sodium peroxide with O_2 at high temperature and pressure

$$Na_2O + O_2 \rightarrow 2NaO_2$$

Structure



The presence of one unpaired electron in 3 electron bond explains paramagnetic character. The superoxide has 17 electrons which give a bond order of 1.5 which occupy the molecular orbitals as shown

$\sigma 1s^2, \sigma^*1s^2, \sigma 2s^2, \sigma^*2s^2, \sigma 2p_x^2, \pi 2p_y^2 = \pi 2p_z^2, \pi^*2p_y^2 = \pi^*2P_z^1$

Stability of oxides : Normal oxide > peroxide > superoxide.

1.3.3 Carbonates and Bicarbonates

Group 1 metals form solid bicarbonates ($MHCO_3$). All alkali metals form carbonates of type M_2CO_3. Due to highly electro positive nature of the alkali metals their carbonates and bicarbonates are highly stable to heat (Li_2CO_3 decomposes easily by heat).

The exceptional behaviour of Li_2CO_3 can be explained by

(a) small size and strong polarisation of Li distorts the e^- cloud of the near by oxygen atom of the large CO_3^{2-} to such an extent that the C—O bond gets weakened.

$$Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$$

(b) Replacement of the larger carbonate ion by a smaller ion leads to increased lattice energy and thus favours the decomposition.

$$M_2CO_3 \xrightarrow{\Delta} M_2O + CO_2\uparrow$$

Na_2CO_3 is used as **washing soda. $NaHCO_3$** is used as **baking soda.** The crystal structure of $NaHCO_3$ and $KHCO_3$ both show hydrogen bonding. In $NaHCO_3$, the HCO_3^- linked into an infinite chain whilst in $KHCO_3$ a dimeric anion is formed.

Reactions

$2HNO_3 + K_2CO_3 \rightarrow 2KNO_3 + CO_2 + H_2O$

$2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + H_2O + CO_2$

$M_2CO_3 + H_2O \rightleftharpoons 2M^+ + HCO_3^- + OH^-$

They hydrolyze to give basic solution.

1.3.4 Halides

All the metals in this group form halides of type MX.

Li^+ is the smallest ion in the group, it would be expected to form hydrated salts more readily than other metals.

Properties :

As evident from their following properties, alkali metal halides are ideal ionic compounds.

(i) All alkali halides except lithium fluoride are freely soluble in water (LiF is soluble in non-polar solvents).

(ii) They have high melting and boiling points.

(a) For the same alkali metal, the melting and boiling points decrease regularly in the order

Fluoride > chloride > bromide > iodide

This is explained on the basis of **lattice energy*** of these metal halides. For the same metal, lattice energy decreases with the decrease in electronegativity of the halogen. For example,

Metal Halide :	NaF	NaCl	NaBr	NaI
Lattice energy (kJ/mole) :	910	769	732	682
Melting point (K) :	1298	1081	1028	934

(b) For the same halide ion, the melting point of lithium halides are lower than those of the sodium halides. However, after sodium the melting points of halides decrease as we move down the group from Na to Cs. This abnormal behaviour shown by lithium halides is probably due to its covalent nature whereas sodium and other halides are ionic in nature. Amongst ionic halides, melting point decreases as lattice energy decreases as we move down the group i.e.,

NaCl >	KC1 >	RbCl >	CsCl
1081 K	1045 K	990 K	918 K

(iii) **Solubility of halides of alkali metals :-** The solubilities of alkali metal halides show a gradation. For example, solubility of alkali metal fluorides in water increases regularly from lithium to caesium.

Metal fluoride :	LiF	NaF	KF	RbF	CsF
Solubility in water at 298 K (gm/litre) :	2-7	42	1020	1310	3700

In case of chlorides, LiCl has much higher solubility in water than NaCl. This is due to small size of Li^+ ion and much higher hydration energy. However, from NaCl to CsCl, solubility in water increases regularly due to decrease in their lattice energy.

(iv) They are good conductors of electricity in the fused state.

(v) They have ionic crystal structure. However, lithium halides have partly covalent character due to polarising power of Li ions.

The structure and stability (solubility) of alkali metal halides are explained by the lattice energy, and polarising power.

(a) Lattice energy :- Lattice energy is the energy released during the formation of a crystal lattice from the respective gaseous cations and anions; or it is the energy required to separate one mole of the solid ionic compound into its gaseous ions. Thus lattice energy (the force of attraction among the ions) is a direct measure of the stability of ionic crystals; higher the lattice energy of a compound lower will be its solubility in water.

When a crystal of an ionic compound comes in contact with a polar solvent such as water, the hydrogen end (positive pole) of the water molecule is attracted to a negative ion while the oxygen end (negative pole) is attracted to a positive ion. This attachment of polar solvent molecules to the ions is known as **solvation (or hydration,** if the solvent is water) of the ions. With the stabilization of the ions by solvation, a large amount of **solvation energy** (or **hydration energy)** is released which if exceeds the lattice energy of the crystal causes the dissolution of the ionic compound in the solvent. On the other hand, if the solvation energy is not enough to counteract the lattice energy, the substance remains insoluble as in case of lithium fluoride. The high lattice energy of lithium fluoride is due to the combination of small lithium ion with small fluoride ion. In general, for a given ion, the lattice energy increases as the size of the oppositely charged ion decreases.

(b) Polarising power and polarisability (Fajan's rule) :- Although an ionic bond in a compound like M^+X^- is considered to be 100% ionic, in some cases (e.g., lithium halides) it is found to have significant covalent character . According to Fajan, when the two oppositely charged ions approach each other, the nature of the bond between them depends upon the effect of one ion on the other.

When two oppositely charged ions approach each other, the positive ion attracts electrons present on the outermost shell of the anion and repels its positively charged nucleus. This results in the distortion, deformation or polarisation of the anion. Thus the power of a cation to distort the anion is known as its polarisation power and the tendency of the anion to get polarised by the cation is known as its **polarisability**. If the polarisation is quite small, an ionic bond is formed, while if the degree of polarisation is large, electrons are drawn from the anion to the cation by electrostatic attraction with the result the electron density between the two ions is increased and the resulting bond becomes covalent in character. In general, greater the polarisation power or polarisability of an ion greater will be its tendency to form covalent bond. Since polarisation power increases with the decrease in the size of the cation while polarisability increases with the increase in the size of the anion, in a compound consisting of large negative ions and small positive ions the polarization may be so marked that the bond becomes covalent. Thus lithium iodide, consisting of Li^+ ions (the smallest alkali metal ion) and I^- ions (the greatest halide ion), is found to be markedly covalent in nature.

Other examples of such ionic-covalent compounds are $AlCl_3$, $FeCl_3$, $SnCl_4$, etc.

Reactions

The alkali metal halides react with the halogen and interhalogen compounds forming ionic polyhalides.

$KI + I_2 \rightarrow K[I_3]$

$KBr + ICl \rightarrow K[Br\,ICl]$

$KF + BrF_3 \rightarrow K[BrF_4]$

1.3.5 Sulphates

They form sulphates of type M_2SO_4

1.4 Anomalous Behaviour of Lithium

Although lithium exhibits most of the characteristic properties of the group I elements, it differs, at the same time, in many respects from them. This anomalous behaviour of lithium is due to extremely small size of lithium atom and its ion. The small size of the Li^+ ion leads to its high charge density. Lithium ion, therefore, possesses the greatest polarising power out of all the alkali metal ions. Hence it exerts a great distorting effect on a negative ion. Consequently, the Li^+ ion has a remarkable tendency towards solvation and covalent bond formation. Further it is important to note that the polarising power of the Li^+ ion is similar to that of Mg^{2+} ion, hence the two elements (placed diagonally in the periodic table) resemble very much in their properties.

(i) Lithium is much harder than the other elements of group I (similarity with Mg which is also a hard metal).

(ii) The m.p. and b.p. of lithium are comparatively high.

(iii) Unlike other elements of this group, lithium is the least reactive as represented by the following points.

- **(a)** Unlike others, it is not affected by air.
- **(b)** Unlike others, it decomposes water very slowly (resemblance with Mg).
- **(c)** Unlike others, it hardly reacts with bromine.
- **(d)** On burning in oxygen, it forms only the monoxide Li_2O, while the others form peroxides (M_2O_2) too. Further, K, Rb and Cs form even the superoxide, MO_2.

(iv) Unlike other elements, it directly combines with nitrogen to form nitride, Li_3N (similarity with Mg):

(v) Lithium is much less electropositive and, therefore, several of its compounds (Li_2CO_3 and LiOH) are less stable (similarity with Mg). For example,

$$2LiOH \rightarrow Li_2O + H_2O$$

$$Mg(OH)_2 \rightarrow MgO + H_2O$$

Hydroxides of other alkali metals sublime unchanged.

(vi) Lithium nitrate, on heating, gives nitrogen dioxide and oxygen leaving behind lithium oxide (similarity with $MgNO_3$), while sodium and potassium nitrates evolve only oxygen, thus leaving nitrites.

$$4LiNO_3 \rightarrow 2Li_2O + 4NO_2 + O_2$$

$$[2Mg(NO_3)_2 \rightarrow 2\,MgO + 4NO_2 + O_2]$$

$$2NaNO_3 \rightarrow 2NaNO_2 + O_2$$

(vii) Most of the lithium salts (e.g., hydroxide, carbonate, oxalate, phosphate and fluoride) are sparingly soluble in water (similarity with Mg.) The corresponding salts of sodium and potassium are freely soluble in water.

(viii) Lithium halides and lithium alkyls are soluble in organic solvents, while those of Na and K are insoluble ; $MgCl_2$ is also soluble in alcohol.

(ix) Lithium chloride (like $MgCl_2$) undergoes hydrolysis in hot water though to a small extent; while NaC*l* and KC*l* do not hydrolyse at all.

(x) Unlike sulphates of other alkali metals, lithium sulphate does not form alums.

(xi) Lithium compounds, particularly, lithium halides are partially covalent in nature. This is due to the tendency of Li^+ to draw electrons towards itself (polarising power). This explains the lower Value (e.g;, 6.25 D in LiI) of the dipole moment of lithium compounds than the expected (e.g., 11.5 D in LiI).

(xii) The ions and its compounds are more heavily hydrated than those of the other alkali metals (similarity with Mg.)

1.5 Extraction of Sodium

Sodium is obtained on large scale by two processes :

1.5.1 Castner's process

In this process, electrolysis of fused sodium hydroxide is carried out at 330ºC using iron as cathode and nickel as anode.

$$2NaOH \rightleftharpoons 2Na^+ + 2OH^-$$

At cathode : $2Na^+ + 2e^- \rightarrow 2Na$

At anode : $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$

During electrolysis, oxygen and water are produced. Water formed at the anode gets partly evaporated and is partly broken down and hydrogen is discharged at cathode.

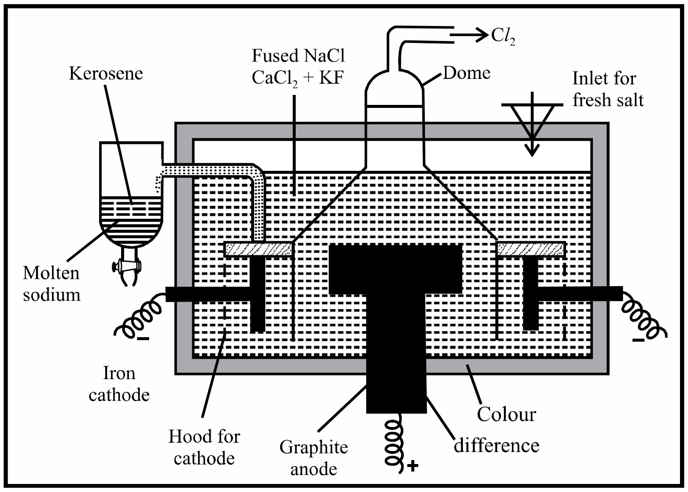
$$H_2O \rightleftharpoons H^+ + OH^-$$

At cathode : $2H^+ + 2e \rightarrow 2H \rightarrow H_2\uparrow$

1.5.2 Down's process

Now-a-days the metal is manufactured by Down's process. It involves the electrolysis of fused sodium chloride containing calcium chloride and potassium fluoride, using iron as cathode and graphite as anode, at about 600º C.

The cell consists of a steel tank lined with heat-resistant bricks. A circular graphite anode is placed in the centre of the cell which is surrounded by a cylindrical iron cathode. The anode and cathode are separated by a steel gauze cylinder through which fused charge can pass. The anode is covered by a dome-shaped steel hood which provides the outlet for the escape of chlorine gas. The molten metal liberated at the cathode moves up and flows into the receiver containing kerosene.



Reactions : $NaCl \rightleftharpoons Na^+ + Cl^-$

At Cathode : $2Na^+ + 2e^- \rightarrow 2Na$

At Anode : $2Cl^- \rightarrow Cl_2 + 2e^-$

Sodium obtained by this method is 99.5% pure.

The electrolysis of pure sodium chloride presents the following difficulties :

(i) The fusion temperature of NaCl is high, i.e., 803ºC (1076 K) which is difficult to maintain.

(ii) Sodium is volatile at this temperature and therefore, a part of it vapourises and forms a metallic fog.

(iii) At this temperature, the products of electrolysis, sodium and chlorine are corrosive and may attack the material of the cell.

To remove the above difficulties, pure sodium chloride is mixed with calcium chloride and potassium fluoride. Calcium chloride and potassium fluoride do not decompose at the voltage employed however, they lower the fusion temperature. The fusion temperature of a mixture containing 40% NaCl and 60% calcium chloride and a very small amount of potassium fluoride becomes about 600°C. The electrolysis of this mixture at 600ºC is done in the electrolytic cell.

EXAMPLE – 1 Alkali metals are paramagnetic but their salts are diamagentic. Explain.

Sol. In metals, the outermost energy shell is singly occupied, but in cations, all the orbitals are doubly occupied (inert gas configuration).

e.g., Na, $1s^2$, $2s^2$ $2p^6$, $3s^2$ $3p^6$, $4s^1$ paramagnetic

Na^+ $1s^2$, $2s^2$ $2p^6$, $3s^2 3p^6$ Diamagnetic

EXAMPLE – 2 Alkali metals are good reducing agents. Explain.

Sol. Alkali metals act as strong reducing agents because they can lose valence electrons readily on account of low ionisation enthalpy values and high values of oxidation potential.

EXAMPLE – 3 Which alkali metal ion has the maximum polarising power and why ?

Sol. Li^+ ion has the maximum polarising power among the alkali metal ions. This is due to small size of Li^+ ion.

EXAMPLE – 4 Li^+ ion is far smaller than other alkali metal ions but it moves through a solution less rapidly than the others. Explain.

Or

The conductance of lithium salts is less in comparison to the salts of other alkali metals. Explain.

Sol. The dense charge of Li^+ attracts several water molecules around it, i.e., Li^+ ion has maximum degree of hydration. Thus, the size of the hydrated lithium ion is largest in comparison to the size of the other alkali metal ions which affects its movement in solution and the conductance is less.

Size : $[Li\,(aq)]^+ > [Na\,(aq)]^+ > [K\,(aq)]^+$

EXAMPLE – 5 Sodium salts in aqueous solutions are either neutral or alkaline in nature. Explain.

Sol. The anions in sodium salts are either from strong acids or weak acids. When anions are from strong acids, there is no hydrolysis and aqueous solutions are neutral (NaCl, $NaNO_3$, Na_2SO_4 solutions are neutral).

On the other hand, when anions are from weak acids, there is hydrolysis and the solutions are alkaline in nature. For example, in the case of sodium carbonate or bicarbonate, solns. are alkaline.

$$CO_3^{2-} + H_2O \rightleftharpoons HCO_3^- + OH^-$$

$$HCO_3^- + H_2O \rightleftharpoons H_2CO_3 + OH^-$$

EXAMPLE – 6 Why do K, Rb and Cs form superoxides in preference to oxides and peroxides on being heated in excess supply of air ?

Sol. K^+, Rb^+ and Cs^+ are large cations in size and superoxide ion (O_2^-) is larger in size in comparison to oxide (O^{2-}) and peroxide (O_2^{2-}) ion. A larger cation can stabilise a large anion and therefore, these metals form superoxides rather than oxides and peroxides.

EXAMPLE – 7 Why is KO_2 paramagnetic ?

Sol. The superoxide O_2^- is paramagnetic because of one unpaired electron in π*2p molecular orbital.

$$KK\,\sigma\,(2s)^2\sigma^*\,(2s)^2\,\sigma\,(2p_x)^2\,\pi\,(2p_x)^2\,(\pi 2p_y)^2\pi^*\,(2p_x)^2\,\pi^*\,(2p_y)^1$$

EXAMPLE – 8 Among the alkali metals which element has :

(i) highest melting point

(ii) highest size of hydrated ion in solution

(iii) strongest reducing agent in solution

(iv) least electronegative

Sol. (i) Li, (ii) $[Li(aq)]^+$ (iii) Li, (iv) Cs

EXAMPLE – 9 What happens when following compounds are heated ?

(a) Li_2CO_3 (b) $Na_2CO_3.10H_2O$

(c) $LiNO_3$ (d) $NaNO_3$

Sol.
$$Li_2CO_3 \longrightarrow Li_2O + CO_2$$

$$Na_2CO_3\,.\,10H_2O \longrightarrow Na_2CO_3 + 10H_2O$$

$$4LiNO_3 \longrightarrow 2Li_2O + 4NO_2 + O_2$$

$$2NaNO_3 \longrightarrow 2NaNO_2 + O_2$$

EXAMPLE – 10 (a) Arrange LiF, NaF, KF, RbF and CsF in order of increasing lattice energy.

(b) Arrange the following in order of the increasing covalent character.

MCl, MBr, MF, MI (where M = alkali metal)

Sol. (a) CsF < RbF < KF < NaF < LiF

(b) MF < MCl < MBr < MI

with increasing size of the anion, covalent character increases.

EXAMPLE – 11 Why a standard solution of sodium hydroxide cannot be prepared by weighing ?

Sol. NaOH is a deliquescent substance. It absorbs moisture and reacts with CO_2 of the atmosphere and both increase its mass. Thus, accurate weighing is difficult.

EXAMPLE – 12 What happens when :

(a) fused sodium reacts with dry ammonia.

(b) sodium hydrogen carbonate is heated.

(c) sodium hydroxide is heated with sulphur ?

Sol. (a) Sodamide is farmed with evolution of hydrogen

$$2Na + 2NH_3 + \rightarrow \underset{\text{Sodamide}}{2NaNH_2} + H_2$$

(b) Sodium carbonate is formed.

$$2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$$

(c) Sodium thiosulphate is formed.

$$4s + 6NaOH \rightarrow \underset{\text{Sodium thiosulphate}}{Na_2S_2O_3} + 2Na_2S + 3H_2O$$

EXAMPLE – 13 Give reasons for the following :

(i) LiCl is more covalent than NaCl

(ii) LiI has lower melting point then LiCl

(iii) $MgCl_2$ is more covalent than NaCl

(iv) CuCl is more covalent than NaCl

Sol. (i) Due to smaller size, Li^+ ion is more polarising than Na^+ and hence LiCl is more covalent than NaCl

(ii) Due to bigger size, I^- is more polarisable than Cl^- and hence LiI is more covalent than LiCl. Therefore, LiI has lower melting point than LiCl.

(iii) Due to higher charge, Mg^{2+} is more polarising than Na^+ and hence $MgCl_2$ is more covalent than NaCl.

(iv) Dut to pseudo inert gas configuration, Cu^+ is more polarising than Na^+, and hence CuCl is more covalent than NaCl.

EXAMPLE – 14 Identify (A), (B), (C) and (D) and give their chemical formulae.

(a) $(A) + NaOH \xrightarrow{\text{Heat}} NaCl + NH_3 + H_2O$

(b) $NH_3 + CO_2 + H_2O \longrightarrow (B)$

(c) $(B) + NaCl \longrightarrow (C) + NH_4Cl$

(d) $(C) \xrightarrow{\text{Heat}} Na_2CO_3 + H_2O + (D)$

Sol. (a) $NH_4Cl + NaOH \xrightarrow{\text{Heat}} NH_3 + NaCl + H_2O$

(A) is ammonium chloride (NH_4Cl).

(b) $NH_3 + CO_2 + H_2O \longrightarrow NH_4HCO_3$

(B) is ammonium bicarbonate (NH_4HCO_3).

(c) $NH_4HCO_3 + NaCl \longrightarrow NaHCO_3 + NH_4Cl$

(C) is sodium bicarbonate ($NaHCO_3$).

(d) $2NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$

(D) is carbon dioxide (CO_2).

EXAMPLE – 15 Arrange the following as specified :

(i) MgO, SrO, K_2O and Cs_2O (increasing order of basic character)

(ii) LiCl, LiBr, LiI (decreasing order of covalent character)

(iii) $NaHCO_3$, $KHCO_3$, $Mg(HCO_3)_2$, $Ca(HCO_3)_2$ (decreasing solubility in water)

(iv) LiF, NaF, RbF, KF and CsF (in order of increasing lattice energy)

(v) Li, Na K (in order to decreasing reducing nature in solution)

Sol. (i) $MgO < SrO < K_2O < Cs_2O$

(ii) LiI > LiBr > LiCl

(iii) $NaHCO_3 < KHCO_3 < Mg(HCO_3)_2 < Ca(HCO_3)_2$

(iv) CsF < RbF < KF < NaF < LiF

(v) Li > K > Na

EXAMPLE – 16 (a) What happens when KO_2 reacts with water ? Give the balanced chemical equation.

(b) Predict giving reason the outcome of the reaction :

$LiI + KF \longrightarrow$

Sol. (a) When KO_2 reacts with water, oxygen is evolved and an alkaline solution containing potassium hydroxide and H_2O_2 is formed ?

$$2KO_2 + 2H_2O \longrightarrow 2KOH + H_2O_2 + O_2$$

(b) LiI reacts with KF i.e., anions are exchanged

$$LiI + KF \longrightarrow LiF + KI$$

The exchange occurs as stable compounds are formed i.e., larger cation stabilizes larger anion and smaller cation stabilizes smaller anion.

2. ALKALINE EARTH METALS

2.1 Introduction

The elements, **beryllium, magnesium, calcium, strontium, barium** and **radium,** constitute Group-II of the periodic table.

All these elements are metals. The oxides of calcium, strontium and barium were known much earlier than the metals themselves and they were called alkaline earths since they were alkaline and existed in the earth. Later, when the elements were discovered, they were named as alkaline earth metals. Radium corresponds to the alkaline earth metals in its chemical properties but, being a radioactive element, it is studied separately along with the other radioactive elements.

2.2 Physical Properties

2.2.1 Atomic size

Atomic size increases down the group.

2.2.2 Oxidation State

The group 2 elements exhibit +2 oxidation state.

2.2.3 Density

The size of group 2 elements are smaller than those of group1 thus these elements have higher density than group 1 elements. Density increases from Be to Ra

Exception : Density of Ca is less than Mg and density of Mg is less than Be.

2.2.4 Nature of bonds

Be mainly forms covalent compound. The rest of the elements in group 2 forms ionic bond.

2.2.5 Hydration energy

The hydration energies of the group 2 ions are four or five time greater than for group 1 ions due to their smaller size and increased charge. $\Delta H_{hydration}$ decreases down the group as the size of the ions increases.

2.2.6 Lattice Energy

Lattice energy of salts of alkali metals having common anion decreases on descending down the group.

2.2.7 Ionization Energy

Since the atoms are smaller than those in group1, the electrons are more tightly held so that the energy required to remove the first electron (first ionization energy) is greater than for group 1. Energy required to remove the second electron is nearly double than the first. Therefore energy required to produce divalent ions for group 2 elements is four times greater than the energy required to produce M^+ from group 1 metals.

2.2.8 Flame Test

When energy is supplied to these elements in a flame, their electrons are excited to higher energy states, as is the case with alkali metals under similar conditions. As the electrons drop back to the original energy level, the extra energy is emitted in the form of visible light with characteristic colours as given below :

Element	Colour
Ca	brick red
Sr	crimson red
Ba	grassy green
Ra	crimson

Beryllium and magnesium atoms are smaller. The electrons in these atoms are, therefore, more strongly bound. Hence these are not excited by the energy of the flame to higher energy states. These elements, therefore, do not give any colour in bunsen flame.

2.2.9 Standard oxidation potential

Standard Oxidation Potential of Alkaline Earth Metals

Element	Oxidation Reaction	Standard Oxidation Potential (volt)
Be	$Be \rightarrow Be^{2+} + 2e^-$	1·85
Mg	$Mg \rightarrow Mg^{2+} + 2e^-$	2·37
Ca	$Ca \rightarrow Ca^{2+} + 2e^-$	2·87
Sr	$Sr \rightarrow Sr^{2+} + 2e^-$	2·89
Ba	$Ba \rightarrow Ba^{2+} + 2e^-$	2·90

2.2.10 Solubility in liquid NH_3

The metals dissolve in liquid ammonia as do group 1 metals. Dilute solutions are blue in colour due to the formation of solvated e^-. The solution decomposes forming amides and evolving H_2

$$2NH_3 + 2e^- \rightarrow 2NH_2^- + H_2$$

2.2.11 Electronegative values

The electronegativity values of group 2 are low but are higher than the values of group 1. The value of decreases down the group.

2.2.12 Colourless and diamagnetism

The elements of alkaline earth metal group form M^{2+} ions, since it does not have any unpaired electron, it is diamagnetic and colourless

2.2.13 Melting and Boiling point

Since the cohesive force decreases down the group the melting point of elements of group 2 decreases down the group.

Exception : Mg has the lowest melting point.

Boiling points do not show regular trends. They are harder than alkali metals .

2.2.14 Metallic properties

Elements of Group-II have typical metallic properties. They show good metallic luster and high electrical as well as thermal conductivity.

GROUP – I & II

OXIDES

Sodium Oxide (Na_2O) :

Preparation :

(i) It is obtained by burning sodium at 180ºC in a limited supply of air or oxygen and distilling off the excess of sodium.

$$2Na + ½O_2 \xrightarrow{180º} Na_2O$$

(ii) By heating sodium peroxide, nitrate or nitrate with sodium.

$$Na_2O_2 + 2Na \longrightarrow 2Na_2O$$

$$2NaNO_3 + 10Na \longrightarrow 6Na_2O + N_2$$

$$2NaNO_2 + 6Na \longrightarrow 4Na_2O + N_2$$

Properties :

(i) It is white amorphous mass.

(ii) It decomposes at 400ºC into sodium peroxide and sodium

$$2Na_2O \xrightarrow{400ºC} Na_2O_2 + 2Na$$

(iii) It dissolve violently in water, yielding caustic soda.

$$Na_2O + H_2O \longrightarrow 2NaOH$$

Sodium Peroxides (Na_2O_2) :

Preparation : It is formed by heating the metal in excess of air or oxygen at 300ºC, which is free from moisture and CO_2.

$$2Na + O_2 \longrightarrow Na_2O_2$$

Properties :

(i) It is a pale yellow solid, becoming white in air from the formation of a film of NaOH and Na_2CO_3.

(ii) In cold water (~0°C) produces H_2O_2 but at room temperature produces O_2. In ice-cold mineral acids also produces H_2O_2.

$$Na_2O_2 + 2H_2O \xrightarrow{\sim 0^\circ C} 2NaOH + H_2O_2$$

$$2Na_2O_2 + 2H_2O \xrightarrow{25^\circ C} 4NaOH + O_2$$

$$Na_2O_2 + H_2SO_4 \xrightarrow{\sim 0^\circ C} Na_2SO_4 + H_2O_2$$

(iii) It reacts with CO_2, giving sodium carbonate and oxygen and hence its use for purifying air in a confined space e.g. submarine, ill-ventilated room,

$$2Na_2O_2 + 2CO_2 \longrightarrow 2Na_2CO_3 + O_2$$

(iv) It is an oxidising agent and oxidises charcoal, CO, NH_3, SO_2.

$3Na_2O_2 + 2C \longrightarrow 2Na_2CO_3 + 2Na$ [deposition of metallic Na]

$$Na_2O_2 + CO \longrightarrow Na_2CO_3$$

$$Na_2O_2 + SO_2 \longrightarrow Na_2SO_4$$

$$3Na_2O_2 + 2NH_3 + \longrightarrow 6NaOH + N_2$$

(v) It contains peroxide ion $[—O—O—]^{-2}$

Uses :

(i) For preparing H_2O_2, O_2

(ii) Oxygenating the air in submarines

(iii) Oxidising agent in the laboratory.

Oxides of Potassium :

	Colours
K_2O	White
K_2O_2	White
K_2O_3	Red
KO_2	Bright Yellow
KO_3	Reddish brown needles

Preparations :

(i) $2KNO_3 + 10K \xrightarrow{\text{heating}} 6K_2O + N_2$

✹ $K_2O \xrightarrow{\text{heating}} K_2O$

(White) (Yellow)

✹ $K_2O + H_2O \longrightarrow 2KOH$

(ii) $2K + O_2 \xrightarrow[\text{air at } 300^\circ C]{\text{Controlled}} K_2O_2$ [Props : Similar with Na_2O_2]

(iii) Passage of O_2 through a blue solution of K in liquid NH_3 yields oxides K_2O_2 (white), K_2O_3(red) and KO_2 (deep yellow) i.e.

$$\text{K in liq. } NH_3 \longrightarrow \underset{\text{white}}{K_2O_2} \longrightarrow \underset{\text{red}}{K_2O_3} \longrightarrow \underset{\text{yellow}}{KO_2}$$

✹ KO_2 reacts with H_2O and produces H_2O_2 and O_2 both

$$2KO_2 + 2H_2O \longrightarrow 2KOH + H_2O_2 + O_2$$

KO_3 : $\underset{\text{(Dry powdered)}}{KOH} + O_3 \text{ (ozonised oxygen)} \xrightarrow{-10^\circ \text{ to } 15^\circ C} \underset{\text{(orange solid)}}{KO_3}$

Magnesium Oxide (MgO) :-

It is also called magnesia and obtained by heating natural magnesite.

$$MgCO_3 \longrightarrow MgO + CO_2$$

Properties :

(i) It is white powder.

(ii) It's m.p. is 2850°C. Hence used in manufacture of refractory bricks for furnaces.

(iii) It is very slightly soluble in water imparting alkaline reaction.

Calcium Oxide (CaO) :

It is commonly called as quick lime or lime and made by decomposing lime stone at a high temperature about 1000°C.

$$CaCO_3 \longrightarrow CaO + CO_2 + 42000 \text{ cal}$$

Properties :

(i) It is white amorphous powder of m.p. 2570°C.

(ii) It emits intense light (lime light), when heated in oxygen-hydrogen flame.

(iii) It is a basic oxide and combines with some acidic oxide e.g.

$$CaO + SiO_2 \longrightarrow CaSiO_3$$

$$CaO + CO_2 \longrightarrow CaCO_3$$

(iv) It combines with water to produce slaked lime.

$$CaO + H_2O \longrightarrow Ca(OH)_2$$

Magnesium Peroxide (MgO_2) and Calcium Peroxide (CaO_2) :-

These are obtained by passing H_2O_2 in a suspension of $Mg(OH)_2$ and $Ca(OH)_2$.

Uses :

MgO_2 is used as an antiseptic in tooth paste and a bleaching agent.

HYDROXIDES

Sodium Hydroxides :

Preparation :

(i) Electrolysis of Brine :

$$NaCl \rightleftharpoons Na^+ + Cl^-$$

At anode : $2Cl^- \longrightarrow Cl_2 + 2e$

At cathode : $Na + e^- \longrightarrow Na$

$$2Na + 2H_2O \longrightarrow 2NaOH + H_2$$

(ii) Caustication of Na_2CO_3 (Gossage's method) :

$$\underset{(suspension)}{Na_2CO_3 + Ca(OH)_2} \rightleftharpoons 2NaOH + CaCO_3$$

Since the K_{sp} $(CaCO_3) < K_{sp}(Ca(OH)_2)$, the reaction shifts towards right.

Properties :

(i) It is white crystalline, deliquescent, highly corrosive solid.

(ii) It is stable towards heat.

(iii) It's aqueous solution is alkaline in nature and soapy in touch.

(iv) $FeCl_3 + 3NaOH \longrightarrow Fe(OH)_3 \downarrow + 3NaCl$

$$NH_4Cl + NaOH \longrightarrow NaCl + NH_3 \uparrow + H_2O$$

$$ZnCl_2 + 2NaOH \longrightarrow Zn(OH)_2 \downarrow + 2NaCl$$

$$Zn(OH)_2 \downarrow + 2NaOH \xrightarrow{Excess} \underset{Soluble}{Na_2ZnO_2} + 2H_2O$$

[Same with $AlCl_3$, $SnCl_2$, $PbCl_2$]

(v) Acidic and amphoteric oxides gets dissolved easily e.g.

$$CO_2 + 2NaOH \longrightarrow Na_2CO_3 + H_2O$$

$$Al_2O_3 + 2NaOH \longrightarrow 2NaAlO_2 + H_2O$$

(vi) Aluminium and Zn metal gives H_2 from NaOH

$$2Al + 2NaOH + 2H_2O \longrightarrow 3H_2 + 2NaAlO_2$$

(vii) Several non metals such as P, S, Cl etc. yield a hydride instead of hydrogen e.g.

$$4P + 3NaOH + 3H_2O \longrightarrow PH_3 + 3NaH_2PO_2$$

Potassium Hydroxide :

Preparation : Electrolysis of aqueous solution of KCl.

Properties : Same as NaOH

** (a) It is stronger base compared to NaOH.

(b) Solubility in water is more compared to NaOH.

(c) In alcohol, NaOH is sparingly soluble but KOH is highly soluble.

(d) As a reagent KOH is less frequently used but in absorption of CO_2, KOH is preferably used compared to NaOH. Because $KHCO_3$ formed is soluble whereas $NaHCO_3$ is insoluble.

Magnesium Hydroxide : It occurs in nature as the mineral brucite.

Preparation : It can be prepared by adding caustic soda solution to a solution of Mg-sulphate or chloride solution.

$$MgSO_4 + 2NaOH \longrightarrow Na_2SO_4 + Mg(OH)_2$$

Properties :

(i) It can be dried at temperature upto 100ºC only otherwise it breaks into its oxide at higher temperature.

$$Mg(OH)_2 \longrightarrow MgO + H_2O$$

(ii) It is slightly soluble in water imparting alkalinity.

(iii) It dissolves in NH_4Cl solution

$$Mg(OH)_2 + 2NH_4Cl \longrightarrow MgCl_2 + 2NH_4OH$$

✹ Thus, $Mg(OH)_2$ is not therefore precipitated from a solution of Mg^{+2} ions by NH_4OH in presence of excess of NH_4Cl.

Calcium Hydroxide :

Preparation : By spraying water on quicklime.

$$CaO + H_2O \longrightarrow Ca(OH)_2$$

Properties :

(i) It is sparingly soluble in water.

(ii) It's solubility in hot water is less than that of cold water. Hence solubility decreases with increase in temperature.

(iii) It readily absorbs CO_2 as used as a test for the gas.

(iv) It is used as a mortar.

[Mortar is a mixture of slaked lime (1 Part) and saved (3 Parts) made into paste with water.]

CARBONATES

Sodium Carbonate :

Preparation :

(i) Leblanc Process :

$$NaCl + H_2SO_4(conc.) \xrightarrow{\text{mild heating}} NaHSO_4 + HCl$$

$$NaCl + NaHSO_4 \xrightarrow[\text{heated}]{\text{Strongly}} \underset{\text{(Salt Cake)}}{Na_2SO_4} + HCl$$

$$Na_2SO_4 + 4C \longrightarrow Na_2S + 4CO \uparrow$$

$$Na_2S + CaCO_3 \longrightarrow Na_2CO_3 + CaS$$

(ii) Solvay Process :

$$NH_3 + H_2O + CO_2 \longrightarrow NH_4HCO_3$$

$$NaCl + NH_4HCO_3 \longrightarrow NaHCO_3 + NH_4Cl$$

$$2NaHCO_3 \xrightarrow{150°C} Na_2CO_3 + H_2O + CO_2$$

Properties :

(i) Anhydrous Na_2CO_3 is called as soda ash, which does not decompose on heating but melts at 852ºC.

(ii) It forms number of hydrates.

(iii) Hydrated Na_2CO_3 is called washing soda ($Na_2CO_3.10\ H_2O$) and is prepared by Le Blanc process solvay process and electrolytic process.

(iii) Na_2CO_3 absorbs CO_2 yielding sparingly soluble sodium bicarbonate which can be calcined at 250º to get pure sodium carbonate.

$$Na_2CO_3 + H_2O + CO_2 \underset{250°C}{\rightleftharpoons} 2NaHCO_3$$

(iv) It dissolved in acid with effervescence of CO_2 and causticised by lime to give caustic soda.

$$Na_2CO_3 + HCl \longrightarrow 2NaCl + H_2O + CO_2$$

$$Na_2CO_3 + Ca(OH)_2 \longrightarrow 2NaOH + CaCO_3$$

Uses : It is widely used in glass making as smelter.

Potassium Carbonate :

By leblanc process, it can be prepared but by solvay process it cannot be prepared because $KHCO_3$ is soluble in water.

Properties : It resembles with Na_2CO_3, m.p. is 900ºC but a mixture of Na_2CO_3 and K_2CO_3 melts at 712ºC.

Uses : It is used in glass manufacturing.

Calcium Carbonate :

It occurs in nature as marble, limestone, chalk, calcite etc. It is prepared by dissolving marble or limestone in HCl and removing iron and aluminium present, by precipitating with NH_3 and then adding $(NH_4)_2\ CO_3$ to the solution.

$$CaCl_2 + (NH_4)_2CO_3 \longrightarrow CaCO_3 + 2NH_4Cl$$

Properties :

(i) It is dissociates above 1000ºC as follows :

$$CaCO_3 \longrightarrow CaO + CO_2$$

(ii) It dissolves in water containing CO_2 forming $Ca(HCO_3)_2$ but is precipitated from the solution by boiling.

$$CaCO_3 \longrightarrow H_2O + CO_2 \underset{\text{boiling}}{\rightleftharpoons} Ca(HCO_3)_2$$

Magnesium Carbonate :

It occurs in nature as magnesite, isomorphous with calcite. It is obtained as a white precipitated by adding sodium bicarbonate to a solution of a magnesium salt but only basic carbonate, called magnesia alba, having the approximate composition $MgCO_3.\ Mg(OH)_2.\ 3H_2O$ is precipitated.

Properties : Same as $CaCO_3$.

BICARBONATES

Sodium bicarbonates :

Preparation : By absorption of CO_2 in Na_2CO_3 solution.

$$Na_2CO_3 + H_2O + CO_2 \rightleftharpoons 2NaHCO_3$$

Uses : It is used in medicine and as baking powder.

Potassium bicarbonates :

Preparation : Same as $NaHCO_3$

Properties : Same as $NaHCO_3$

But it is more alkaline and more soluble in water compared to $NaHCO_3$.

Magnesium bicarbonate :

$$MgCO_3 + CO_2 + H_2O \underset{}{\overset{\text{boiling}}{\rightleftharpoons}} Mg(HCO_3)_2$$

Calcium bicarbonate :

$$CaCO_3 + CO_2 + H_2O \rightleftharpoons Ca(HCO_3)_2$$

CHLORIDES

Sodium Chloride : Prepared from brine containing 25% NaCl.

Properties :

(i) It is nonhygroscopic but the presence of $MgCl_2$ in common salt renders it hygroscopic.

(ii) It is used to prepare freezing mixture in laboratory [Ice-common salt mixture is called freezing mixture and temperature goes down to –23°C.]

(iii) For melting ice and snow on road.

Potassium Chloride : It also occurs in nature as sylvite (KCl) or carnallite $KCl . MgCl_2 . 6H_2O$.

Uses : It is used as fertiliser.

Magnesium Chloride :

Preparation : By dissolving $MgCO_3$ in dil. HCl.

$$MgCO_3 + 2HCl \longrightarrow MgCl_2 + H_2O + CO_2$$

Properties :

(i) It crystallises as hexahydrate. $MgCl_2 . 6H_2O$.

(ii) It is deliquescent solid.

(iii) This hydrate undergoes hydrolysis as follows :

$$MgCl_2 + H_2O \longrightarrow Mg(OH)Cl + HCl$$

$$Mg(OH)Cl \longrightarrow MgO + HCl$$

- Hence, anhy. $MgCl_2$ cannot be prepared by heating this hydrate.
- Because of the formation of HCl sea water cannot be used in marine boilers which corrodes the iron body.

(iv) Anhydrous $MgCl_2$ can be prepared by heating a double salt like. $MgClO_2 . NH_4Cl. 6H_2O$ as follows :

$$MgCl_2 . NH_4Cl_2 . 6H_2O \xrightarrow[\Delta]{-H_2O} MgCl_2NH_4Cl$$

$$\xrightarrow[\Delta]{\text{strong}} MgCl_2 + NH_3 + HCl$$

Sorel Cement : It is a mixture of MgO and $MgCl_2$ (paste like) which sets to hard mass on standing. This is used in dental filling, flooring etc.

Calcium Chloride :

(i) It is the by-product in solvay process.

(ii) It may also be prepared by dissolving the carbonate in HCl

$$CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$$

Properties :

(i) It is deliquescent crystals.

(ii) It gets hydrolysed like $MgCl_2$ hence anhydrous $CaCl_2$ cannot be prepared.

$$CaCl_2 + H_2O \rightleftharpoons CaO + 2HCl$$

Hence, anhy. $CaCl_2$ is prepared by heating $CaCl_2 . 6H_2O$ in a current of HCl (dry).

(iii) Anhy. $CaCl_2$ is used in drying gases and organic compounds but not NH_3 or alcohol due to the formation of $CaCl_2 . 8NH_3$ and $CaCl_2 . 4C_2H_5OH$.

SULPHATES

Sodium Sulphate :

Preparation :

It is formed in the I^st step of leblanc process by heating common salt with sulphuric acid.

$$2NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl$$

Thus the salt cake formed is crystallised out from its aqueous solution as $Na_2SO_4 . 10H_2O$. This is called as Glauber's salt.

- One interesting feature of the solubility of glauber's salt is ; when crystallised at below 32.4ºC, then $Na_2SO_4 \cdot 10H_2O$ is obtained but above 32.4ºC, Na_2SO_4 (anhy.) comes out.

Properties : it is reduced to Na_2S when fused with carbon.

$$Na_2SO_4 + 4C \longrightarrow Na_2S + 4CO$$

Uses : It is used in medicine.

Potassium Sulphate :

It occurs in stassfurt potash beds as schonite K_2SO_4, $MgSO_4 \cdot 6H_2O$ and Kainite, $KCl \cdot MgSO_4 \cdot 3H_2O$ from which it is obtained by solution in water and crystallisation. It separates from the solution as crystals whereas Na_2SO_4 comes as decahydrate.

Uses : It is used to prepare alum.

Magnesium Sulphate :

Preparation :

(i) It is obtained by dissolving kieserite $MgSO_4 \cdot H_2O$ in boiling water and then crystallising the solution as a hepta hydrate. i.e. $MgSO_4 \cdot 7H_2O$. It is called as **Epsom salt.**

(ii) It is also obtained by dissolving magnesite in hot dil. H_2SO_4.

$$MgCO_3 + H_2SO_4 \longrightarrow MgSO_4 + H_2O + CO_2$$

or by dissolving dolomite ($CaCO_3 \cdot MgCO_3$) in hot dil. H_2SO_4 and removing the insoluble $CaSO_4$ by filtration.

(iii) It is isomorphous with $FeSO_4 \cdot 7H_2O$, $ZnSO_4 \cdot 7H_2O$

Calcium Sulphate : It occurs as anhydrite $CaSO_4$ and as the dihydrate $CaSO_4 \cdot 2H_2O$, gypsum, alabaster or satin-spar.

Properties :

(i) Gypsum ($CaSO_4 \cdot 2H_2O$) $\underset{+H_2O}{\overset{120°C}{\rightleftharpoons}}$ $2CaSO_4 \cdot H_2O$ (Plaster of paris)

$$2CaSO_4 \cdot H_2O \xrightarrow{200°C} CaSO_4 \text{ (anhydrous)}$$

Dead burnt plaster

(ii) Solubility of $CaSO_4$ at first increases upto a certain point and then decreases with rise of temperature.

(iii) Plaster of paris is used in wood making due to its porous body.

EXAMPLE – 17
(a) Mg_3N_2 when reacted with water, gives off NH_3 but HCl is not obtained from $MgCl_2$ on reaction with water at room temperature. Why ?
(b) The crystalline salts of alkaline earth metals contain more water of crystallization than corresponding alkali metal salts. Why ?

Sol. (a) Mg_3N_2 is a salt of a strong base and weak acid (NH_3), hence its hydrolysis is possible.

$$Mg_3N_2 + 6H_2O \longrightarrow 3Mg(OH)_2 + 2NH_3$$

$MgCl_2$ is a salt of a strong base, [$Mg(OH)_2$], and a strong acid, (HCl), and hence, its hydrolysis is not possible.

(b) Due to small size and high nuclear charge, alkaline earth metal ions have higher tendency of hydration in comparison to alkali metal ions. Thus, the salts of alkaline earth metals contain more water of crystallisation than the salts of alkali metals.

EXAMPLE – 18 What happens when :
(i) Beryllium carbide reacts with water.
(ii) Magnesium nitrate is heated.
(iii) Quick lime is heated in electric furnace with powdered coke.
(iv) NaOH solution is added to $ZnCl_2$ solution.

Sol. (i) Methane gas is evolved.

$$Be_2C + 4H_2O \longrightarrow 2Be(OH)_2 + CH_4$$

(ii) Brown coloured gas, NO_2, is evolved.

$$2Mg\,NO_3 \longrightarrow 2MgO + 4NO_2 + O_2$$

(iii) Calcium carbide is formed with evolution of CCl.

$$CaO + 3C \xrightarrow[\text{Arc}]{\text{Electric}} CaC_2 + CO$$

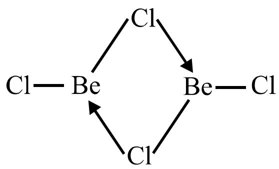
(iv) While precipitate of Zn $(OH)_2$ is formed which dissolves in excess of NaOH forming sodium zincate.

$$ZnCl_2 + 2NaOH \longrightarrow \underset{\text{ppt.}}{Zn(OH)_2} + 2NaCl$$

$$Zn(OH)_2 + 2NaOH \longrightarrow \underset{\text{Soluble}}{Na_2ZnO_2} + 2H_2O$$

EXAMPLE – 19 Draw the structure of (i) $BeCl_2$ (vapour state) (ii) $BeCl_2$ (solid).

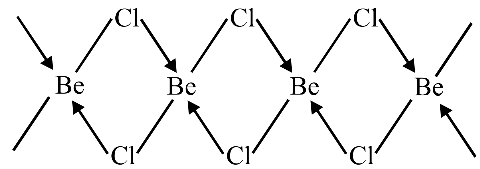
Sol. In vapour state, it has chlorobridged dimer structure which dissociates into linear monomer at 1000º C.



Dimer

Monomer

In the solid phase, it has a polymeric structure with chlorobridges in which a halogen atom bonded to one beryllium atom uses a lone pair of electrons to form a coordinate bond and to another atom by a covalent bond.



EXERCISE – 1

(Basic Level Objective Questions)

1. Sodium metal cannot be stored under

(a) benzene (b) kerosene oil

(c) ethanol (d) toluene

2. Chile saltpetre is and Indian salt petre are !

(a) $NaNO_3$, KNO_3 (b) $Ba(NO_3)_2$, $Cu(NO_3)_2$

(c) $NaNO_2$, $NaNO_3$ (d) KNO_3, $NaNO_3$

3. Baking powder has one of the following constituents, which is

(a) Na_2CO_3 (b) Na_2SO_4

(c) $KHCO_3$ (d) $NaHCO_3$

4. Which of the following oxides is formed when potassium metal is burnt in excess of air ?

(a) K_2O (b) KO

(c) K_2O_2 (d) KO_2

5. When CO_2 is bubbled into an aqueous solution of Na_2CO_3 the compound formed is

(a) NaOH (b) $NaHCO_3$

(c) H_2O (d) OH^-

6. Which of the following reacts with water at a highest rate ?

(a) Na (b) Li

(c) K (d) Rb

7. Which of the following metals cannot form nitride with nitrogen ?

(a) Li (b) Na

(c) Mg (d) Ca

8. A solution of sodium metal with liquid ammonia is strongly reducing due to the presence of

(a) free ions (b) amide ion

(c) solvated electrons (d) None of these

9. Alkali metals present in their compounds are always

(a) monovalent (b) bivalent

(c) trivalent (d) in free state

10. When sodium reacts with sufficient amount of air the product obtained is

(a) Na_2O (b) Na_2O_2

(c) NaO_2 (d) NaO

11. The metal that dissolved in ammonia to give a blue colour is

(a) Sn (b) Pb

(c) Zn (d) Na

12. Which of the following properties is not true for an alkali metal ?

(a) Low atomic volume (b) Low ionization energy

(c) Low density (d) Low electronegativity

13. Sodium conducts electricity because

(a) it is soluble in water

(b) it has only one electron in the outermost orbit

(c) it has mobile electrons

(d) it is an alkali metal

14. When K_2O is added to water the solution is basic because it contains a significant concentration of

(a) oxide ions (b) peroxide ions

(c) hydroxide ions (d) superoxide ions

15. Excess of Na^+ ions in human system causes

(a) high blood pressure (b) anaemia

(c) low blood pressure (d) None of these

16. Which of the following has lowest melting point ?

(a) Li (b) Na

(c) K (d) Cs

17. Which of the following alkali metal hydroxides is the strongest base ?

(a) LiOH (b) KOH

(c) NaOH (d) RbOH

18. Leblanc process is employed in the manufacture of

(a) baking soda (b) washing soda

(c) KC*l* (d) plaster of Paris

19. Which of the following is plaster of Paris ?

(a) $CaSO_4 . 2H_2O$ (b) $(CuSO_4)_2 \cdot \frac{1}{2} H_2O$

(c) $(CaSO_4)_2 \cdot H_2O$ (d) $CaSO_4$

20. Temporary hardness may be removed from water by adding

(a) $Ca(OH)_2$ (b) $CaCO_3$

(c) $CaSO_4$ (d) HCl

21. K_2CO_3 cannot be prepared by Solvay process because

(a) K_2CO_3 decomposed in the process

(b) $KHCO_3$ is soluble in water

(c) $KHCO_3$ is insoluble in water

(d) $KHCO_3$ is thermally stable

22. Which of the following on heating gives oxides quickly ?

(a) Na_2CO_3 (b) K_2CO_3

(c) Li_2CO_3 (d) Rb_2CO_3

23. Which of the following compounds is not soluble in water ?

(a) LiF (b) NaCl

(c) NaBr (d) LiBr

24. Which of the following metals forms ionic carbide when heated with carbon ?

(a) Na (b) K

(c) Li (d) Cs

25. Which of the following hydrides does not burn in air even on strong heating ?

(a) CaH_2 (b) NaH

(c) LiH (d) SrH_2

26. Which of the following compounds gives CO_2 gas on decomposition ?

(a) Li_2CO_3 (b) $MgCO_3$

(c) Na_2CO_3 (d) Both (a) and (b)

27. Which of the following pairs of elements cannot form alloys ?

(a) K, Cs (b) Cs, Rb

(c) Na, K (d) Li, Rb

28. Which of the following elements does not form peroxide ?

(a) Sr (b) Na

(c) Be (d) Ba

29. Which of the following is nitrolim ?

(a) $CaCN_2$ (b) $CaCN_2 + C$

(c) $Ca(OH)_2 + (CN)_2$ (d) $CaCN_2 + CO$

30. The dilute solution of alkaline earth metals in liquid ammonia shows

(a) a bright blue colour due to formation of metal cluster

(b) a bright blue colour due to solvated electrons

(c) a bronze colour due to solvated electrons

(d) a bronze colour due to formation of metal cluster

31. Which of the following is called milk of magnesia ?

(a) Suspension of $Mg(OH)_2$

(b) Solution of $MgSO_4$

(c) Suspension of Mg

(d) Solution of $Mg(HCO_3)_2$

32. Beryllium hydroxide on reaction with nitric acid forms

(a) hydrated Be $(NO_3)_2$ (b) anhydrous Be $(NO_3)_2$

(c) $[Be_4O(NO_3)_6]$ (d) None of these

33. Which of the following is incorrect ?

(a) Both $BeCl_2$ and $AlCl_3$ have bridged chloride structure

(b) Both $BeCl_2$ and $AlCl_3$ are strong Lewis acids

(c) Both $BeCl_2$ and $AlCl_3$ are covalent compounds

(d) $BeCl_2$ is weak Lewis acid while $AlCl_3$ is strong Lewis acid

34. Which of these is dead burnt plaster ?

(a) $CaSO_4$ (b) $CaSO_4 . 2H_2O$

(c) $(CaSO_4)_2 . H_2O$ (d) CaO

35. Epsom salt is

(a) $MgSO_4 . 7H_2O$ (b) $CaSO_4 . H_2O$

(c) $MgSO_4 . 2H_2O$ (d) $BaSO_4 . 2H_2O$

36. A mixture of cement clinker and gypsum is called

(a) portland cement (b) limestone

(c) mortar (d) None of these

37. Which of the following metals has most stable carbonate ?

(a) Na (b) Mg

(c) Al (d) Si

38. Which of the following has the lowest melting point ?

(a) LiCl (b) NaCl

(c) KCl (d) RbCl

39. The correct order of stability of hydrides is

(a) $CaH_2 > SrH_2 > BaH_2$ (b) $CaH_2 > SrH_2 = BaH_2$

(c) $CaH_2 < SrH_2 < BaH_2$ (d) $CaH_2 < SrH_2 > BaH_2$

40. Lithium on reaction with ammonia gives

(a) Li_3N (b) Li_2NH

(c) $LiNO_2$ (d) $LiNO_3$

41. Based on lattice energy and other considerations which one of the following alkali metal chlorides is expected to have the highest melting point ?

(a) KCl (b) RbCl

(c) LiCl (d) NaCl

42. Use of potassium superoxide (KO_2) in submarine and spacecraft cylinder is

(a) to absorb moisture (b) to generate heat

(c) to absorb CO_2 and increase O_2

(d) to produce CO_2

43. The difference of water molecules in gypsum and plaster of Paris is

(a) $\frac{5}{2}$ (b) 2

(c) $\frac{1}{2}$ (d) $1\frac{1}{2}$

44. Which is the strongest reducing agent ?

(a) Mg (b) Rb

(c) Na (d) K

45. The decomposition temperature is maximum for

(a) $MgCO_3$ (b) $SrCO_3$

(c) $CaCO_3$ (d) $BaCO_3$

46. On strong heating, sodium bicarbonate changes into

(a) sodium monoxide (b) sodium hydroxide

(c) sodium carbonate (d) sodium peroxide

47. Which is most basic in character ?

(a) CsOH (b) KOH

(c) NaOH (d) LiOH

48. The metal which is the best reducing agent in aqueous solution is :

(a) Na (b) K

(c) Rb (d) Li

49. The alkali metals dissolve in liq. NH_3 to give a blue solution which is conducting due to :

(a) ammoniated electrons (b) loss of electrons

(c) ammoniated cations (d) both (a) and (c)

50. The decrease in melting and boiling points of alkali metals with rise in atomic number is due to :

(a) the weakening of covalent bonding

(b) weak ionic bonding

(c) Van der Wall's forces

(d) the weakening of metallic bonding.

51. Which of the following alkali metals emits light of the largest wavelength in the flame test ?

(a) Na (b) Li

(c) K (d) Cs

52. Ionic mobility of Li^+ is less than that of Na^+ and K^+ ions both, because :

(a) Ionisation potential of Li is small

(b) Charge density of Li^+ ion is high

(c) Hydration tendency of Li^+ ion is high

(d) Li^+ has two electrons

53. Pure anhydrous mangnesium chloride ($MgCl_2$) can be prepared from hydrated salt ($MgCl_2 \cdot 6H_2O$) by :

(a) heating the hydrate to red heat in the atmosphere of HCl gas

(b) melting the hydrate

(c) heating the hydrate with coke

(d) heating the hydrate with Mg ribbon.

54. Which of the following is least soluble in water ?

(a) $CaCO_3$ (b) $NaHCO_3$

(c) Na_2SO_4 (d) $KHCO_3$

55. $BeSO_4$ is soluble in water but $BaSO_4$ is insoluble, because :

(a) $BeSO_4$ is ionic and $BaSO_4$ is covalent.

(b) $BeSO_4$ is crystalline while $BaSO_4$ is amorphous.

(c) $BeSO_4$ has smaller lattice energy and high heat of hydration as compared to $BaSO_4$.

(d) None of these.

56. The sulphate of which metal has the highest solubility in water ?

(a) Ca (b) Ba

(c) Sr (d) Mg

57. The metal that is extracted from sea water is :

(a) Fe (b) Ca

(c) Mg (d) Au

58. Alkaline earth metals form M^{2+} ions instead of M^+ ions because :

(a) M^{2+} ions carry more charge

(b) their IE_2 values are not different from their IE_1 values.

(c) M^+ ions donot have stable configuration

(d) M^{2+} ions have more hydration energy.

59. Washing soda has formula -

(a) $Na_2CO_3 7H_2O$ (b) $Na_2CO_3 . 10H_2O$

(c) $Na_2CO_3 3H_2O$ (d) Na_2CO_3

60. Which of the following metals is present in chlorophyll-

(a) Mg (b) Be

(c) Ca (d) None

61. Sodium reacts with water more vigorously than lithium because -

(a) It has higher atomic weight

(b) It is more electronegative

(c) It is more electropositive

(d) It is a metal

62. Cs^+ ions impart violet colour to Bunsen are flame. This is due to the fact that the emitted radiations of

(a) high energy (b) lower frequencies

(c) longer wave-lengths (d) zero wave number

63. An alkaline earth metal (M) gives a salt with chlorine, which is soluble in water at room temperature. It also forms an insoluble sulphate whose mixture with a sulphide of a transition metal is called 'lithopone' -a white pigment. Metal M is

(a) Ca (b) Mg

(c) Ba (d) Sr

64. The reaction of an element A with water produces combustible gas B and an aqueous solution of C. When another substance D reacts with this solution C also produces the same gas B.D also produces the same gas even on reaction with dilute H_2SO_4 at room temperature. Element A imparts golden yellow colour to Bunsen flame. Then A, B, C and D may be identified as

(a) Na, H_2, NaOH and Zn (b) K, H_2, KOH and Zn

(c) K, H_2,NaOH and Zn (d) Ca, H_2, $CaCOH_2$ and Zn

65. The hydroxide of alkaline earth metal, which has the lowest value of solubility product (K_{sp}) at normal temperature (25ºC) is

(a) $Ca(OH)_2$ (b) $Mg(OH)_2$

(c) $Sr(OH)_2$ (d) $Be(OH)_2$

66. (Yellow ppt) $T \xleftarrow{K_2CrO_4/H^+} X \xleftarrow{\text{dil.HCl}}$ Y(Yellow ppt) + Z.↑ (pungent smelling gas)

If X gives green flame test. Then, X is

(a) $MgSO_4$ (b) BaS_2O_3

(c) $CuSO_4$ (d) PbS_2O_3

67. The golden yellow colour associated with NaCl to Bunsen flame can be explained on the basis of

(a) low ionisation potential of sodium

(b) emission specturm

(c) photosensitivity of sodium

(d) sublimation of metallic sodium of yellow vapours

68. The salt which finds uses in qualitative inorganic analysis is

(a) $CuSO_4{\cdot}5H_2O$ or $ZnSO_4{\cdot}5H_2O$

(b) $K_2SO_4{\cdot}Al_2(SO_4)_3{\cdot}24H_2O$

(c) $Na(NH_4) HPO_4{\cdot}4H_2O$

(d) $FeSO_4{\cdot}(NH_4)_2SO_4{\cdot}6H_2O$

69. Fire extinguishers contain

(a) conc. H_2SO_4 solution

(b) H_2SO_4 and $NaHCO_3$ solutions

(c) $NaHCO_3$ solution

(d) $CaCO_3$ solution

70. $CsBr_3$ contains

(a) Cs–Br covalent bonds (b) Cs^{3+} and Br^- ions

(c) Cs+ and Br_3^- ions (d) Cs^{3+} and Br_3^{3-} ions

71. EDTA is used in the estimation of

(a) Mg^{2+} ions (b) Ca^{2+} ions

(c) both Ca^{2+} and Mg^{2+} ions

(d) Mg^{2+} ions but not Ca^{2+} ions

72. aq. $NaOH + P_4$ (white) $\rightarrow PH_3 + X$; compound X is

(a) NaH_2PO_2 (b) $NaHPO_4$

(c) Na_2CO_3 (d) $NaHCO_3$

73. $Y \xleftarrow{\Delta,205°C} CaSO_4.2H_2O \xrightarrow{\Delta,120°C} X$. X and Y are respectively

(a) plaster of paris, dead burnt plaster

(b) dead burnt plaster, plaster of paris

(c) CaO and plaster of paris

(d) plaster of paris, mixture of gases

74. A metal M readily forms water soluble sulphate, and water insoluble hydroxide $M(OH)_2$. Its oxide MO is amphoteric, hard and having high melting point. The alkaline earth metal M must be

(a) Mg (b) Be

(c) Ca (d) Sr

75. When K_2O is added to water, the solution becomes basic in nature because it contains a significant concentration of

(a) K^+ (b) O^{2-}

(c) OH^- (d) O_2^{2-}

76. (Milky Cloud) $C \xleftarrow{CO_2} A + Na_2CO_3 \rightarrow B + C$

The chemical formulae of A and B are

(a) NaOH and $Ca(OH)_2$ (b) $Ca(OH)_2$ and NaOH

(c) NaOH and CaO (d) CaO and $Ca(OH)_2$

77. An aqueous solution of an halogen salt of potassium reacts with same halogen X_2, to give KX_3, a brown coloured solution, in which halogen exists as X_3^- ion, X_2 as a Lewis acid and X^- as a Lewis base, halogen X is

(a) chlorine (b) bromine

(c) iodine (d) fluorine

78. The correct order of basic-strength of oxides of alkaline earth metals is

(a) BeO > MgO > CaO > SrO

(b) SrO > CaO > MgO > BeO

(c) BeO > CaO > MgO > SrO

(d) SrO > MgO > CaO > BeO

79. NaOH (Solid) + CO $\xrightarrow{200°C}$ X; product X is

(a) $NaHCO_3$ (b) Na_2CO_3

(c) HCOONa (d) H_2CO_3

80. $X \xrightarrow{N_2,\Delta} Y \xrightarrow{H_2O} Z$ (colourless gas) $\xrightarrow{CuSO_4} T$ (blue colour)

Then, substances Y and T are

(a) $Y = Mg_3N_2$ and $T = CuSO_4.5H_2O$

(b) $Y = Mg_3N_2$ and $T = CuSO_4.4NH_3$

(c) $Y = Mg(NO_3)_2$ and $T = CuO$

(d) $Y = MgO$ and $T = CuSO_4.4NH_3$

81. Which of the following substances (s) is/are used in laboratory for drying purposes ?

(a) anhydrous P_2O_5 (b) graphite

(c) anhydrous $CaCl_2$ (d) Na_3PO_4

82. Nitrogen dioxide cannot be prepared by heating

(a) KNO_3 (b) $AgNO_3$

(c) $Pb(NO_3)_2$ (d) $Cu(NO_3)_2$

83. In $LiAlH_4$, metal Al is present in

(a) anionic part (b) cationic part

(c) in both anionic and cationic part

(d) neither in cationic nor in anionic part

84. $X \xrightarrow{CoCl_2} CaCl_2 + Y\uparrow$; the effective ingredient of X is

(a) OCl^- (b) Cl^-

(c) OCl^+ (d) OCl_2^-

85. $X + C + Cl_2 \xrightarrow[\text{of about 1000K}]{\text{High temperature}} Y + CO; Y + 2H_2O \rightarrow Z + 2HCl$

Compound Y is found in polymeric chain structure and is an electron deficient molecule. Y must be

(a) BeO (b) $BeCl_2$

(c) BeH_2 (d) $AlCl_3$

86. $MgBr_2$ and MgI_2 are soluble in acetone because of

(a) Their ionic nature (b) Their coordinate nature

(c) Their metallic nature (d) Their covalent nature

87. Which of the following is not the characteristic of barium ?

(a) It emits electrons on exposure to light

(b) It is a silvery white metal

(c) It forms $Ba(NO_3)_2$ which is used in preparation of green fire

(d) Its ionization potential is lower than radium.

88. Which is not correctly matched ?

(1) Basic strength of oxides	$Cs_2O < Rb_2O < K_2O < Na_2O < Li_2O$
(2) Stability of peroxides	$Na_2O_2 < K_2O_2 < Rb_2O_2 < Cs_2O_2$
(3) Stability of bicarbonates	$LiHCO_3 < NaHCO_3 < KHCO_3 < RbHCO_3 < CsHCO_3$
(4) Melting point	$NaF < NaCl < NaBr < NaI$

(a) 1 and 4 (b) 1 and 3

(c) 1 and 2 (d) 2 and 3

$$NaNO_3 \xrightleftharpoons{500°C} A(s) + B(g)$$

$$NaNO_3 \xrightleftharpoons{800°C} C(s) + B(g) + D(g)$$

89. Compound A is produced by absorbing dinitrogen trioxide in Na_2CO_3 solution. Compound A is :

(a) Na_2O (b) $NaNO_2$

(c) N_2O (d) Na_2O_2

90. Gas B is paramagnetic and support the combustion. Compound B is :

(a) N_2 (b) N_2O

(c) O_2 (d) Na_2O

91. D is a inert gas, which is also obtained by strongly heating the ammonium nitrite. Compound D is :

(a) N_2O (b) NH_3

(c) O_2 (d) N_2

EXERCISE - 2

(Basic Level Subjective Questions)

1. Why are alkali metals not found in nature ?

2. Explain why can alkali and alkaline earth metals not be obtained by chemical reduction methods ?

3. Beryllium and magnesium do not give colour to flame whereas other alkaline earth metals do so. Why ?

4. Potassium carbonate cannot be prepared by Solvay process. Why ?

5. What happens when (i) magnesium is burnt in air (ii) quick lime is heated with silica (iii) chlorine reacts with slaked lime (iv) calcium nitrate is heated ?

6. Draw the structure of (i) $BeCl_2$ (vapour) (ii) $BeCl_2$ (solid).

7. Why are lithium salts commonly hydrated and those of the other alkali ions usually anhydrous ?

8. What happens when

(i) sodium metal is dropped in water ?

(ii) sodium metal is heated in free supply of air ?

(iii) sodium peroxide dissolves in water ?

9. Comment on each of the following observations :

(a) The mobilities of the alkali metal ions in aqueous solution are $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$

(b) Lithium is the only alkali metal to form a nitride directly.

10. State as to why

(a) a solution of Na_2CO_3 is alkaline ?

(b) alkali metals are prepared by electrolysis of their fused chlorides ?

(c) sodium is found to be more useful than potassium ?

11. Explain why halides of beryllium fume in moist air but other alkaline earth metal halides do not.

12. Why superoxides of alkali metals are paramagnetic while normal oxides are diamagnetic ?

13. Water is a liquid, while H_2S is a gas at ordinary temperature. Explain.

14. Sodium carbonate is prepared by Solvay process but the same process is not extended to the manufacture of potassium carbonate, explain.

15. Give reasons in one or two sentences for the following

"H_2O_2 is a better oxidising agent than H_2O."

16. Arrange the following increasing order of basic strength :

$$MgO, SrO, K_2O, NiO, Cs_2O$$

17. Complete and balance the following chemical reaction

Anhydrous potassium nitrate is heated with excess of metallic potassium

$$KNO_3\,(s) + K\,(s) \rightarrow .... + ...$$

18. The crystalline salts of alkaline earth metals contain more water of crystallization than the corresponding alkali metal salts. Why ?

19. Hydrogen peroxide acts both as an oxidizing and as a reducing agent in alkaline solution towards certain first row transition metal ions. Illustrate both these properties of H_2O_2 using chemical equations.

20. Give reasons for the following in one or two sentences only :

"$BeCl_2$ can be easily hydrolysed."

21. Identify (X) in the following synthetic scheme and write their structures.

$$Ba\overset{*}{C}O_3 + H_2SO_4 \longrightarrow X(\text{gas})\ (C^* \text{ denotes } C^{14})$$

EXERCISE - 3

(JEE Mains & JEE Advanced Corner)

JEE Advanced Question

1. Gypsum is **(1978)**

(a) $MgSO_4 \cdot 7H_2O$ (b) $CuSO_4 \cdot 5H_2O$

(c) $CaSO_4 \cdot \frac{1}{2} H_2O$ (d) $CaSO_4 \cdot 2H_2O$

2. A black sulphide is formed by the action of H_2S on **(1978)**

(a) Sodium chloride (b) Cadmium chloride

(c) Zinc chloride (d) Cupric chloride

3. The temporary hardness of water is due to calcium bicarbonate can be removed by adding **(1979)**

(a) $CaCO_3$ (b) $Ca(OH)_2$

(c) $CaCl_2$ (d) HCl

4. Calcium is obtained by **(1980)**

(a) electrolysis of molten $CaCl_2$

(b) electrolysis of solution of $CaCl_2$ in water

(c) reduction of $CaCl_2$ with carbon

(d) roasting of limestone

5. HCl is added to the following oxides. Which one would give H_2O ? **(1980)**

(a) MnO_2 (b) PbO_2

(c) BaO (d) NO_2

6. A solution of sodium metal in liquid ammonia is strongly reducing due to the presence of **(1981)**

(a) sodium atoms (b) sodium hydride

(c) sodium amide (d) solvated electrons

7. Heavy water is **(1983)**

(a) H_2O^{18}

(b) water obtained by repeated distillation

(c) D_2O

(d) water at 4°C

8. A solution of sodium sulphate in water is electrolysed using inert electrodes. The products at cathode and anode are respectively **(1984)**

(a) H_2, O_2 (b) O_2, H_2

(c) O_2, Na (d) O_2, SO_2

9. The hydration energy of Mg^{++} is larger than that of : **(1984)**

(a) Al^{3+} (b) Na^+

(c) Be^{++} (d) Mg^{3+}

10. Molecular formula of Glauber's salt is : **(1985)**

(a) $MgSO_4.7H_2O$ (b) $CuSO_4.5H_2O$

(c) $FeSO_4.7H_2O$ (d) $Na_2SO_4.10H_2O$

11. The pair of compounds which cannot exist together in solution is : **(1986)**

(a) $NaHCO_3$ and NaOH (b) Na_2CO_3 and $NaHCO_3$

(c) Na_2CO_3 and NaOH (d) $NaHCO_3$ and NaCl

12. The metallic lustre exhibited by sodium is explained by **(1987)**

(a) diffusion of sodium ions

(b) oscillation of loose electrons

(c) excitation of free protons

(d) existence of body centered cubic lattice

13. The species that do not contain peroxide ions, is **(1992)**

(a) PbO_2 (b) H_2O_2

(c) SrO_2 (d) BaO_2

14. A dilute aqueous solution of Na_2SO_4 is electrolysed using platinum electrodes. The products at the anode and cathode are respectively **(1993)**

(a) O_2, H_2 (b) $S_2O_8^{2-}$, Na

(b) O_2, Na (d) $S_2O_8^{2-}, H_2$

15. Fire extinguishers contain H_2SO_4 and : **(1994)**

(a) $NaHCO_3$ and Na_2CO_3 (b) $NaHCO_3$ solution

(c) Na_2CO_3 (d) $CaCO_3$

16. NO_2 is obtained by heating : **(1994)**

(a) $CsNO_3$ (b) KNO_3

(c) $LiNO_3$ (d) $NaNO_3$

17. The following compounds have been arranged in order of their increasing thermal stabilities. Identify the correct order. **(1996)**

K_2CO_3 (I) $MgCO_3$ (II) $CaCO_3$ (III) $BeCO_3$ (IV)

(a) I < II < III < IV (b) IV < II < III < I

(c) IV < II < I < III (d) II < IV < III < I

18. Sodium nitrate decomposes above 800°C and does not give : **(1998)**

(a) N_2 (b) O_2

(c) NO_2 (d) Na_2O

19. Highly pure dilute solution of sodium in liquid ammonia : **(1998)**

(a) shows blue colour

(b) do not exhibit electrical conductivity

(c) produces sodium amide

(d) produces hydrogen gas

20. The set representing the correct order of first ionization potential is **(2001)**

(a) K > Na > Li (b) Be > Mg > Ca

(c) B > C > N (d) Ge > Si > C

21. A sodium salt on treatment with $MgCl_2$ gives white precipitate only on heating. The anion of the sodium salt is **(2004)**

(a) HCO_3^- (b) CO_3^{2-}

(c) NO_3^- (d) SO_4^{2-}

22. $MgSO_4$ on reaction with NH_4OH and Na_2HPO_4 forms a white crystalline precipitate. What is its formula ? **(2006)**

(a) $Mg(NH_4)PO_4$ (b) $Mg_3(PO_4)_2$

(c) $MgCl_2 . MgSO_4$ (d) $MgSO_4$

23. The compound (s) formed upon combustion of sodium metal in excess air is (are) **(2009)**

(a) Na_2O_2 (b) Na_2O

(c) NaO_2 (d) NaOH

24. The compound (s) formed upon combustion of sodium metal in excess air is (are) **(2009)**

(a) Na_2O_2 (b) Na_2O

(c) NaO_2 (d) NaOH

JEE Mains Question

25. The metallic sodium dissolves in liquid ammonia to form a deep blue coloured solution. The deep blue colour is due to formation of : **(2002)**

(a) solvated electron, $e(NH_3)_x^-$

(b) solvated atomic sodium, $Na(NH_3)_y$

(c) $(Na^+ + Na^-)$ (d) $NaNH_2 + H_2$

26. A metal M readily forms its sulphate MSO_4 which is water-soluble. It forms its oxide MO which becomes inert on heating. It forms an insoluble hydroxide $M(OH)_2$ which is soluble in NaOH solution. Then M is **(2002)**

(a) Mg (b) Ba

(c) Ca (d) Be.

27. KO_2 (potassium superoxide) is used in oxygen cylinder in space and submarines because it **(2002)**

(a) absorbs CO_2 and increases O_2 content

(b) Eliminate moisture

(c) absorbs CO_2 (d) Produces ozone

28. The substance not likely to contain $CaCO_3$ is **(2003)**

(a) calcined gypsum (b) sea shells

(c) dolomite (d) a marble statue

29. The solubilities of carbonates decrease down the magnesium group due to a decrease in **(2003)**

(a) Inter ionic attraction

(b) Entropy of solution formation

(c) Lattice energies of solids

(d) Hydration energies of cations

30. Beryllium and aluminium exhibit many properties which are similar. But, the two elements differ in **(2004)**

(a) exhibiting maximum covalency in compounds

(b) exhibiting amphoteric nature in their oxides

(c) forming covalent halides

(d) forming polymeric hydrides

31. Based on lattice energy and other considerations which one of the following alkali metal chlorides is expected to have the highest melting point ? **(2005)**

(a) RbCl (b) KCl

(c) NaCl (d) LiCl

32. The ionic mobility of alkali metal ions is aqueous solution is maximum for **(2006)**

(a) Li^+ (b) Na^+

(c) K^+ (d) Rb^+

EXERCISE – 4

(Mind Bogglers)

1. Why does the following reaction :

$$>C-Cl + MF \longrightarrow\ >C-F + MCl$$

proceed better with KF than with NaF ?

2. The enthalpy of formation of hypothetical CaCl (s) is theoretically found to be – 188 kJ mol^{-1} and ΔH_f^o for $CaCl_2$ (s) is – 795 kJ mol^{-1}. Calculate ΔH_f^o for the following disproportionating reaction.

$2CaCl\,(s) \rightarrow CaCl_2\,(s) + Ca(s)$

3. On the basis of the following values of hydration energies and lattice energies, calculate heat of solution of LiCl, NaCl and KCl.

Hydration Energy $\Delta H_{hydr.}$ **(kj mol^{-1})**		**Lattice Energy** ΔH_U **(kJ mol^{-1})**	
Li^+	– 499	LiCl	– 840
Na^+	– 390	NaCl	– 776
K^+	– 305	KCl	– 703
Cl^-	– 382		

4. Calcium burns in nitrogen to produce a white powder which dissolves in sufficient water to produce a gas A and an alkaline solution. The solution on exposure to air produces a thin solid layer of B on the surface. Identify the compounds A and B.

5. Element A burns in nitrogen to give an ionic compound B. Compound B reacts with water to give C and D. A solution of C becomes 'milky' on bubbling carbon dioxide gas. Identify A, B, C and D.

6. A white solid is either Na_2O or Na_2O_2. A piece of red litmus paper turns white when it is dipped into a freshly made aqueous solution of the white solid.

(i) Identify the substance and explain with balanced equation.

(ii) Explain what would happen to the red litmus if the white solid were the other compound.

ANSWER KEY

Excercise - 1

1. (c)	2. (a)	3. (d)	4. (d)	5. (b)	6. (b)	7. (b)	8. (c)	9. (a)	10. (b)
11. (d)	12. (a)	13. (c)	14. (c)	15. (a)	16. (d)	17. (d)	18. (b)	19. (c)	20. (a)
21. (b)	22. (c)	23. (a)	24. (c)	25. (c)	26. (d)	27. (d)	28. (c)	29. (b)	30. (b)
31. (a)	32. (c)	33. (d)	34. (a)	35. (a)	36. (a)	37. (a)	38. (a)	39. (a)	40. (b)
41. (d)	42. (c)	43. (d)	44. (b)	45. (d)	46. (c)	47. (a)	48. (d)	49. (d)	50. (d)
51. (b)	52. (c)	53. (a)	54. (a)	55. (c)	56. (d)	57. (c)	58. (d)	59. (b)	60. (a)
61. (c)	62. (a)	63. (c)	64. (a)	65. (d)	66. (b)	67. (a)	68. (c)	69. (b)	70. (c)
71. (c)	72. (a)	73. (a)	74. (b)	75. (c)	76. (b)	77. (c)	78. (b)	79. (c)	80. (b)
81. (a, c)	82. (a)	83. (a)	84. (a)	85. (b)	86. (d)	87. (a)	88. (b)	89. (b)	90 (c)
91 (d)									

Excercise - 3

1. (d)	2. (d)	3. (b)	4. (a)	5. (None)	6. (d)	7. (c)	8. (a)	9. (b)	10. (d)
11. (a)	12. (b)	13. (a)	14. (a)	15. (a)	16. (c)	17. (b)	18. (a)	19. (a)	20. (b)
21. (a)	22. (a)	23. (a, b)	24. (a, b)	25. (a)	26. (d)	27. (a)	28. (a)	29. (d)	30. (a)
31. (c)	32. (d)								

Excercise - 4

2. – 419 kJ

3. ΔH_{sol} (LiCl) = – 41 kJ mol^{-1}, ΔH_{sol} (NaCl) = + 4 kJ mol^{-1} , ΔH_{sol} (Kcl) = + 16 kJ mol^{-1}

Dream on !!